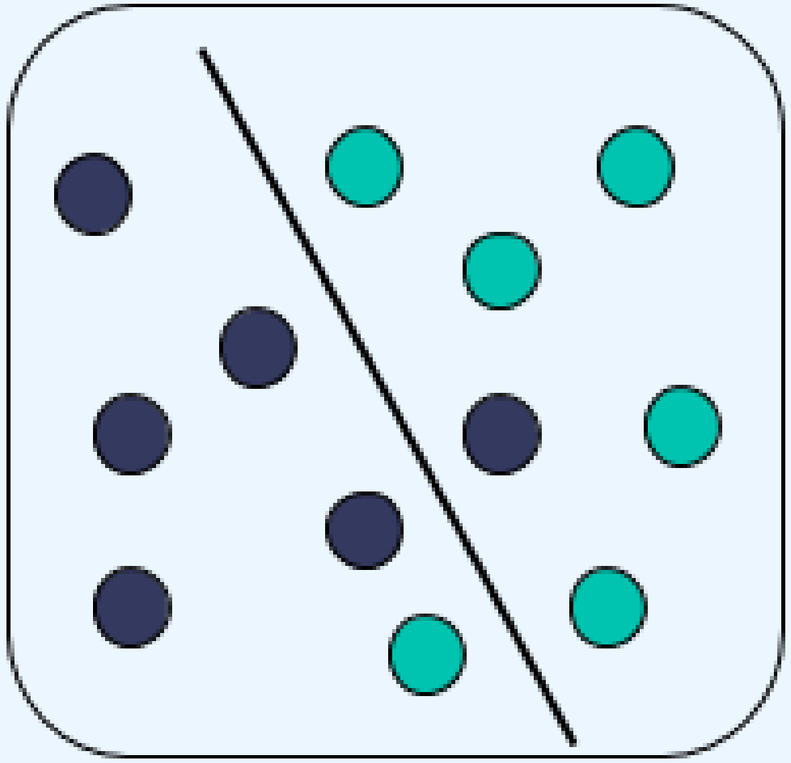
AIGP

Model Types

Linear and statistical



by Evispot

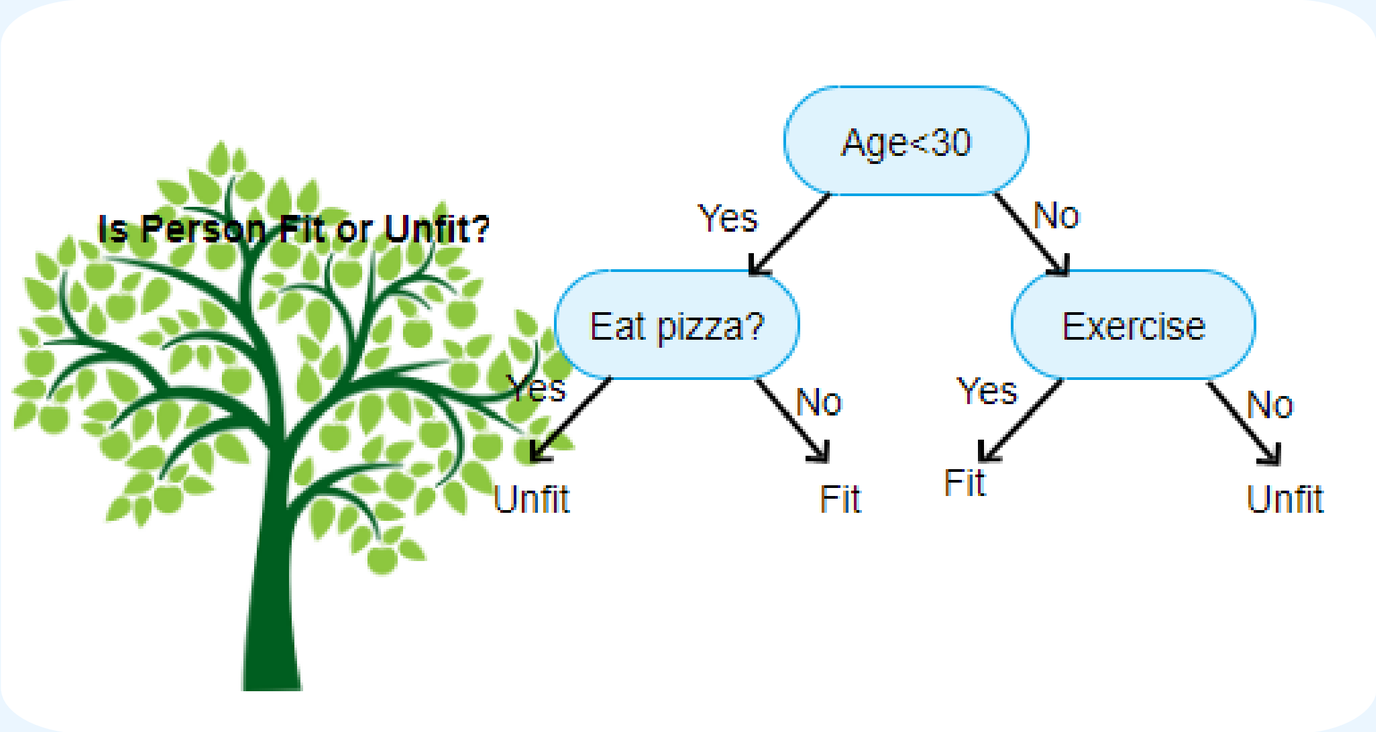
- Linear models predict a dependent variable using one or more independent variables through a linear equation.
- How traffic is related to ad?
- + predictable

- Statistical models represent data using various methods like regression, hypothesis testing, and probability distributions. These models help in making predictions and understanding data patterns.

- Usage:
- Predicting stock prices based on historical data and market trends.
- Forecasting patient outcomes based on medical history and treatment plans.
- Segmenting customers based on purchasing behavior to tailor marketing strategies.

Decision Tree

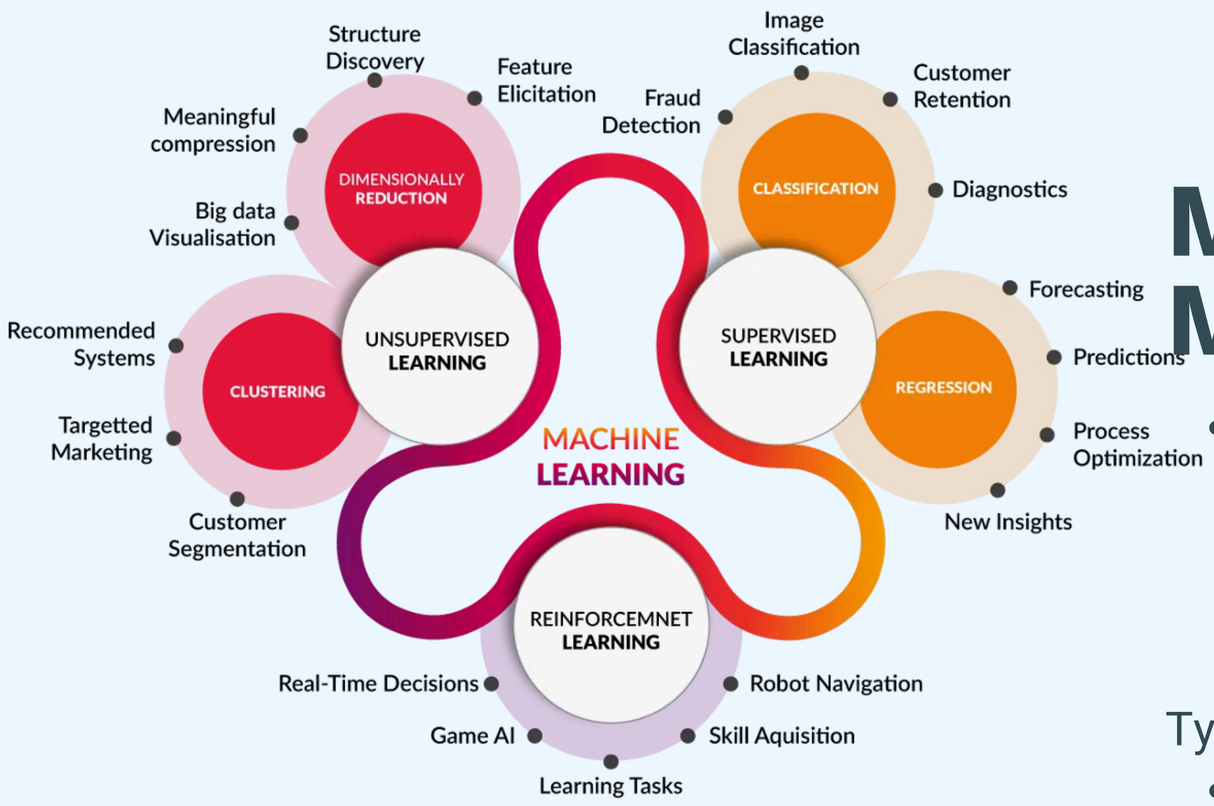
- are a type of supervised learning algorithm used for classification and regression tasks. They split data into subsets based on the value of input features, creating a tree-like model of decisions.



Benefits:

- Easy to understand and interpret.
- Can handle both numerical and categorical data.
- Non-parametric, meaning they do not assume a specific distribution of data.

by Tom Thomas



Machine Learning Models

- Machine learning models use algorithms to learn from and make predictions on data. They include a wide range of techniques, from simple linear regression to complex neural networks.

Types of Models:

- Supervised Learning: Algorithms learn from labeled data (e.g., classification, regression).
- Unsupervised Learning: Algorithms identify patterns in unlabeled data (e.g., clustering, association).
- Semi-Supervised Learning: Combines labeled and unlabeled data for training.
- Reinforcement Learning: Algorithms learn by interacting with an environment to maximize cumulative reward.

by Deepika Yadov

Technopedia

Neural Networks

- Neural networks are a subset of machine learning models inspired by the human brain. They consist of layers of interconnected nodes (neurons) that process data.
-

Types of Neural Networks:

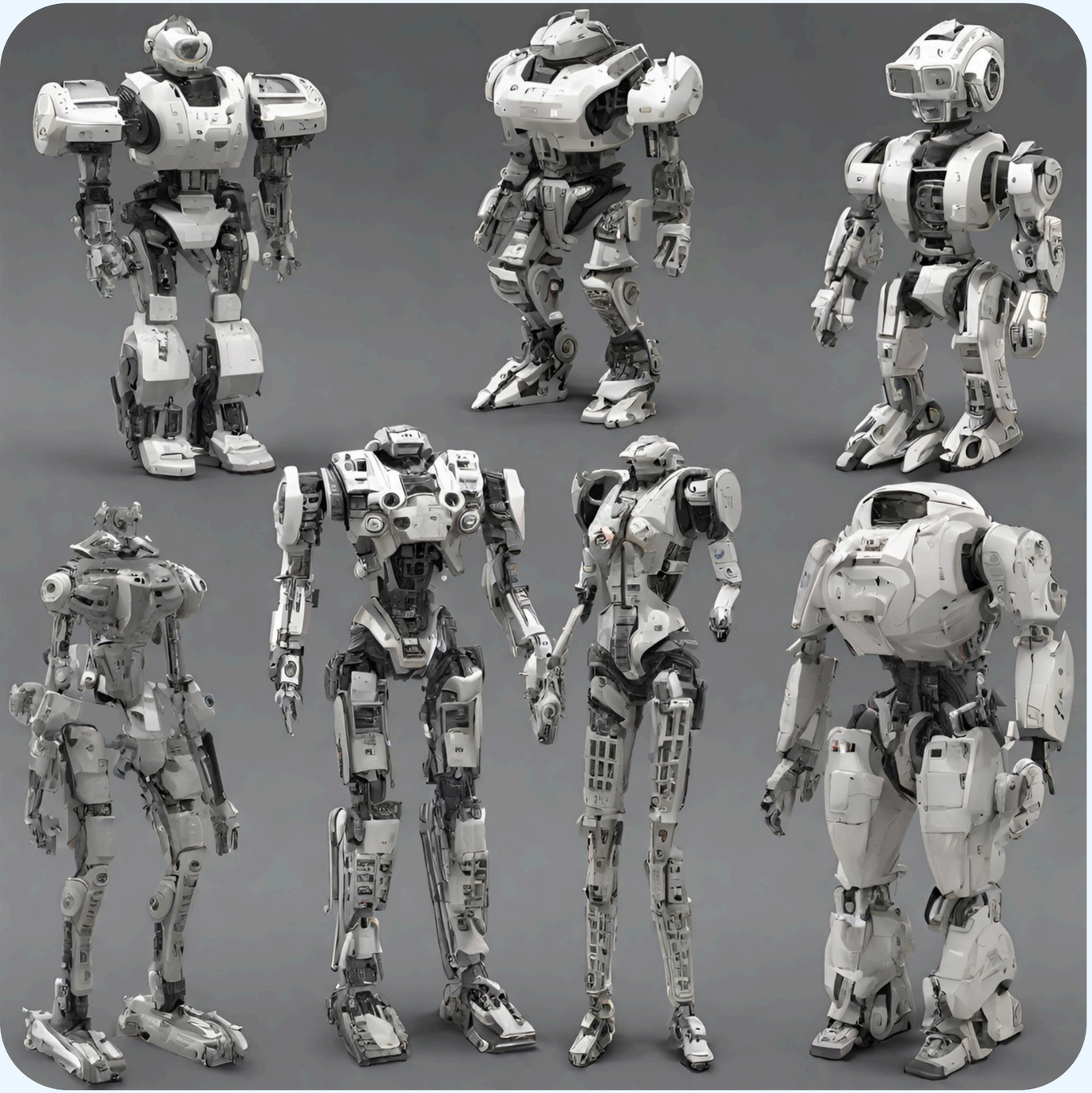
- Feedforward Neural Networks- Data moves in one direction from input to output.
- Convolutional Neural Networks (CNNs)- Specialized for processing grid-like data such as images.
- Recurrent Neural Networks (RNNs)- Designed for sequential data like time series or text.

Language models

Reinforcement models

computer vision

Speech recognition



Robotics

- Robotics involves the design and use of robots, which are programmable machines capable of performing tasks autonomously or semi-autonomously.